

Yene Irvine

yeneirvine@gmail.com • yeneirvine.com • github.com/yirvine • linkedin.com/in/yene-irvine

Education & Certifications

M.ENG., SOFTWARE ENGINEERING

University of Calgary, Calgary AB
GPA of 3.94; graduated May 2024

B.ENG., CHEMICAL ENGINEERING

Dalhousie University, Halifax, NS
GPA of 3.80; graduated April 2018

AWS CERTIFIED CLOUD PRACTITIONER CLF-C02, 2024

MICROSOFT CLOUD CERTIFIED: AZURE FUNDAMENTALS AZ-900, 2024

Work Experience

DATA ENGINEER | Fuel Transport Inc. | Montreal, QC | September 2024 – Present

- Built and integrated a Snowflake Rate Editor (FastAPI, ReactJS) to enable real-time editing of company-wide client rates, powering 100s of daily API calls and contributing to \$1.5M+ in revenue over 5 months
- Developed and implemented a custom Azure Data Factory pipeline to migrate data from IBM DB2 to Snowflake, replacing Fivetran (a managed ELT platform) and saving ~\$50K annually in third-party tool costs
- Deployed backend microservices on Azure Container Apps to handle and parse high-volume XML-based SOAP API traffic and dynamic rate updates across enterprise clients

CONTROLS ENGINEER | MCA Consultants | Charlottetown, PE & Halifax, NS | Apr. 2019 – Jan. 2022

Worked on 2 large-scale expansion projects w/ a pharmaceutical manufacturing company. My key contributions:

- Developed and commissioned 25 complex control narratives for utilities and process control systems, collaborating closely with PLCs, process equipment, contractors to enhance BV Building Automation System
- Developed a program integrated into Microsoft Excel to determine relieving requirements for pressurized vessels based on NFPA and ASME standards; this program was integral in designing venting configurations and specifying required relieving equipment for 50+ major pressurized process vessels
- Led a team to commission all process and utility systems and associated equipment under budget

Projects

- **Detecting Breathing Irregularities with Deep Learning:** Built a 3D-CNN with University of Calgary Biometrics Lab to detect sleep-related medical emergencies via thermal video, using HPC for compute and WandB for real-time tracking at the Calgary Drop-In Centre
- **RuntIt:** a mobile running app using JavaScript (React Native) and NestJS, integrating real-time GPS tracking, Firebase authentication, and social features to enhance user engagement
- **DJ Set Lister:** a cloud-based web application that leverages Spotify's web API and custom-built optimization algorithms to dynamically generate and optimize DIY-DJ set playlists, hosted on AWS EC2

Technical Skills & Languages

- **Frontend, Backend Development:** TypeScript, JavaScript (React), Node.js, FastAPI, SQL, Snowflake
- **Cloud Platforms & DevOps:** AWS (Certified), Azure (Certified, incl. Functions, Container Apps, Azure Data Factory), strong transferable knowledge to Google Cloud Platform, Docker, Terraform, GitHub Actions
- **Data Analysis & ML:** Python (NumPy, PyTorch), parsing and communicating XML and JSON data
- **Other Skills:** Proven leadership, project management, and effective team communication in both technical and non-technical environments